



The Internet of Things: Connecting Our World

Welcome to EmAIoT Academy

Meet Your Instructor: Omar Sameh

Introducing Omar Sameh, a Senior Software Engineer with extensive expertise in AI and IoT technologies. His qualifications include:

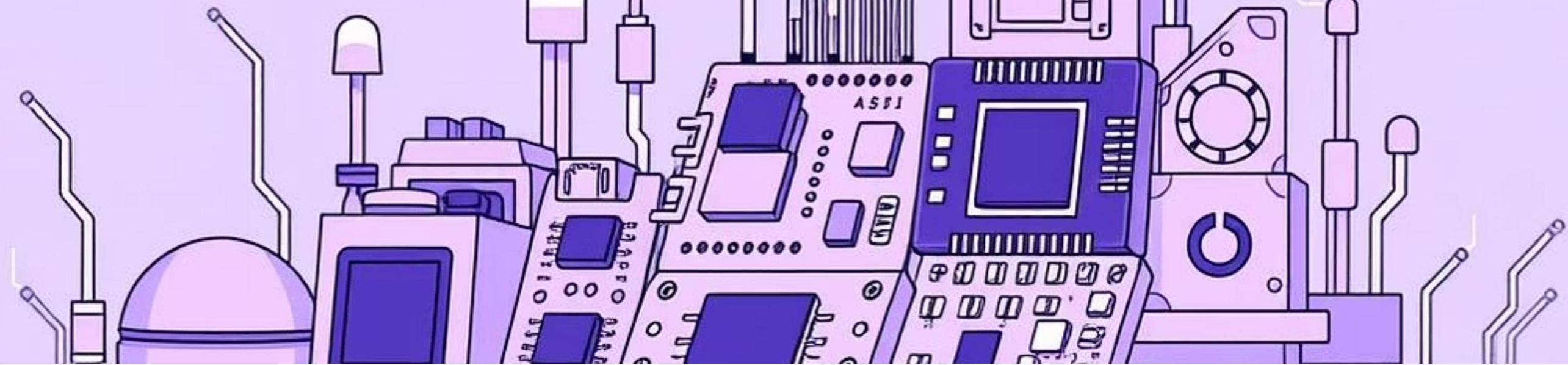
- Senior Software Engineer (current position)
- NTI DEY-Program Certification in "AI & IoT"
- CISCO Networking Academy Certification in IoT
- Participant in 2nd IEEE Global Conference on IoT
- Participant in 5G Opportunities and Challenges Workshop at NTI





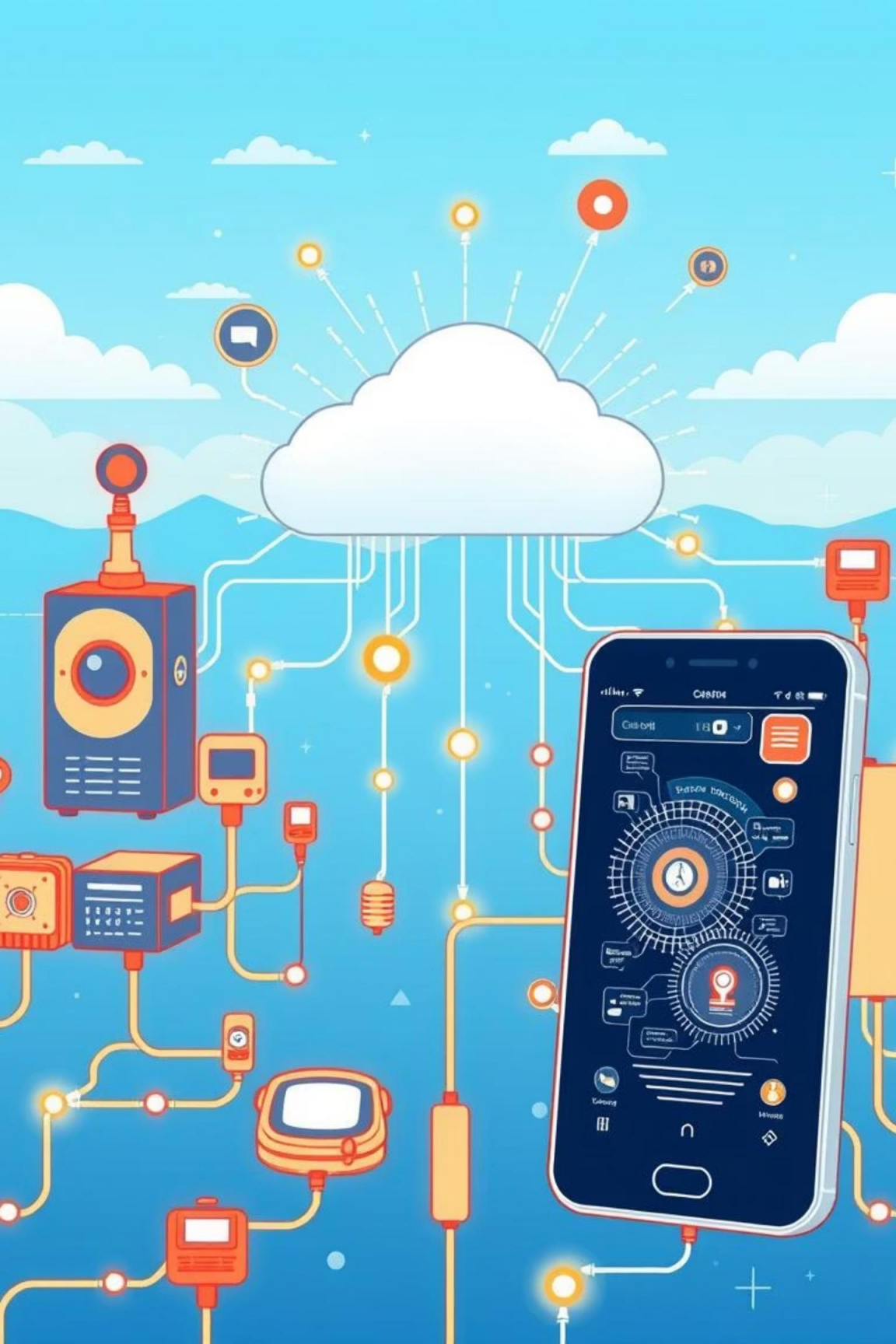
Internet of Things

- Connecting everyday objects to the internet.
- Collecting and exchanging data.
- Transforming industries and daily life.



Building Blocks of IoT

- Sensors and Actuators
- Connectivity (Networks)
- Data Processing & Cloud Platforms



IoT Stack



Smart Devices

- Sensors and actuators
- Collect data
- Physical layer



Connectivity

- Network protocols
- Data transmission
- Bridge devices to cloud



The Cloud

- Data storage & processing
- Backend infrastructure
- Scalable solutions

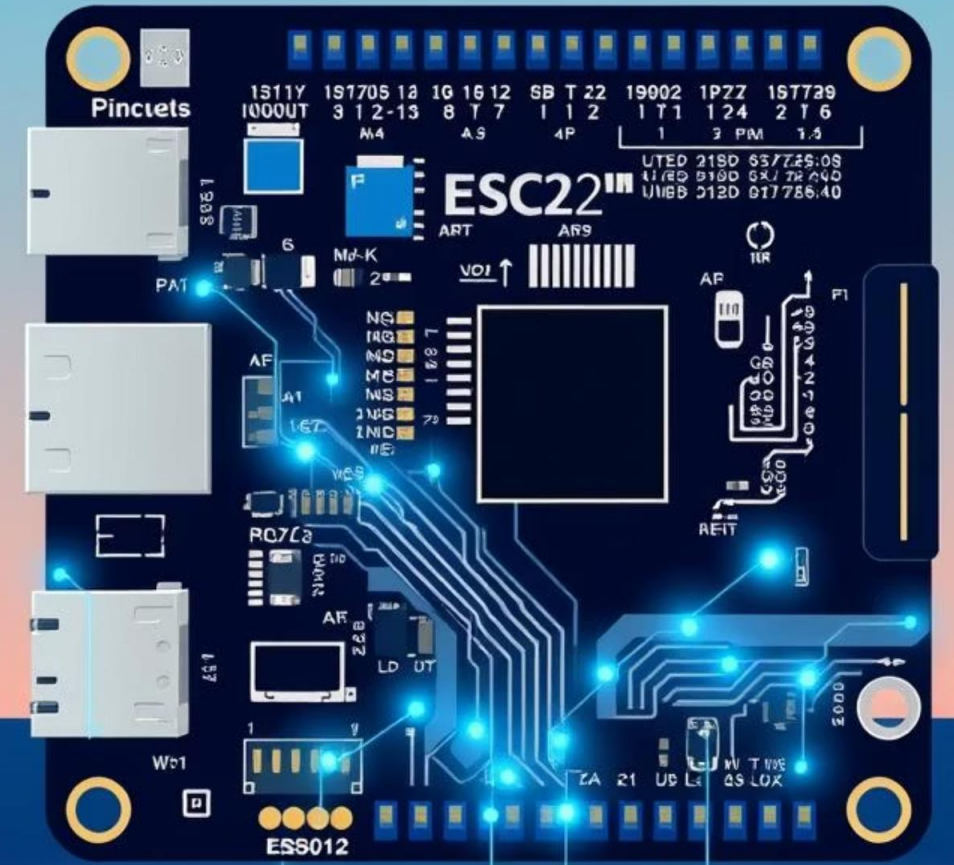


Smart Apps

- User interfaces
- Insights & control
- Actionable outcomes

IoT Hardware Essentials

- Microcontrollers & processors
- Sensors and actuators
- Connectivity modules



How Devices Talk to Each Other

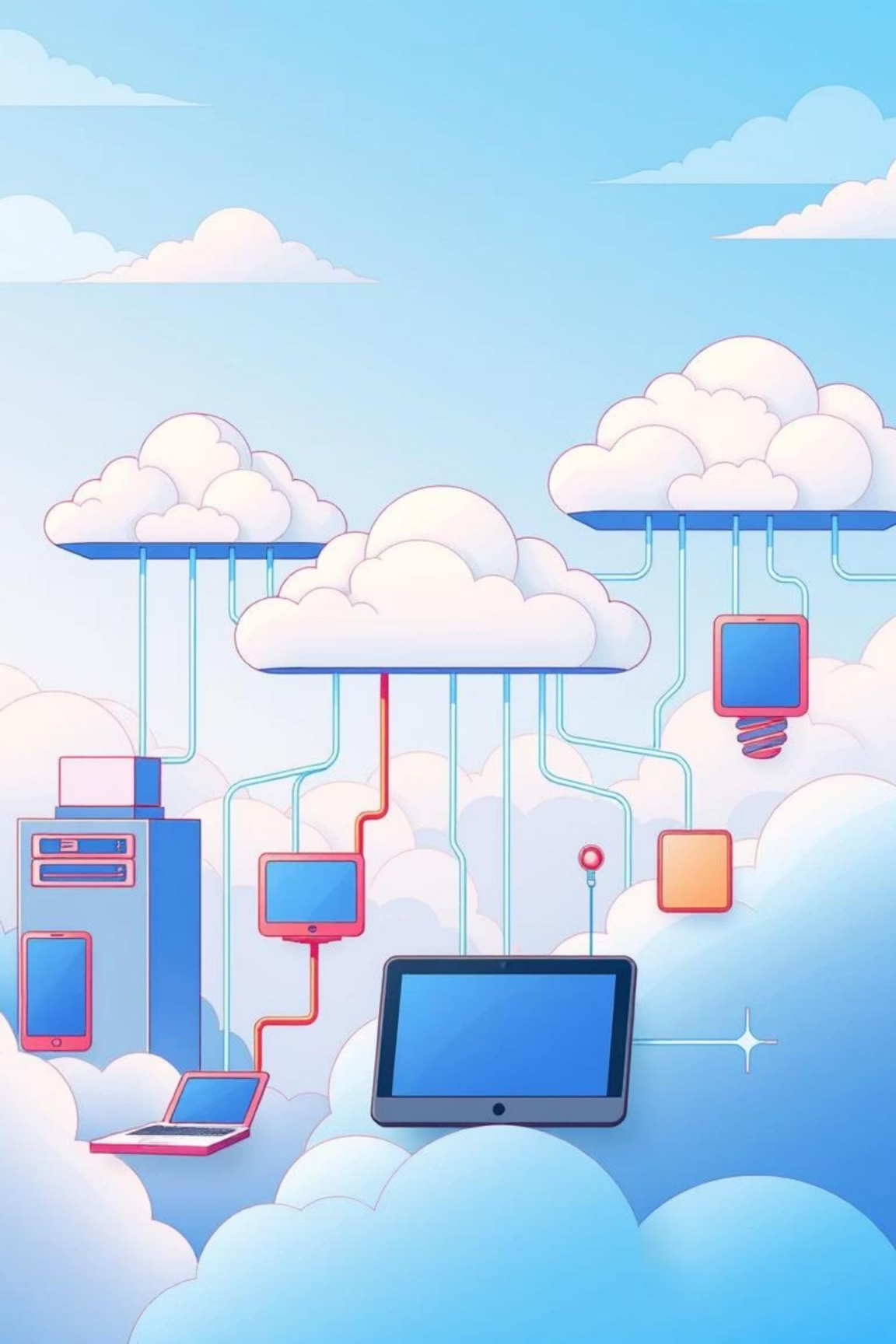
- Understand common IoT communication protocols
- Explore wireless technologies like Wi-Fi, Bluetooth, and Cellular
- Learn about data exchange and network security



IoT Security

- Protecting IoT devices from cyber threats.
- Ensuring data privacy and integrity.
- Best practices for secure IoT implementation.





Cloud Platforms for IoT

- Seamless device connectivity
- Scalable data management
- Real-time analytics & insights

IoT Data Management

- Collecting data from IoT devices
- Storing and processing large data volumes
- Extracting insights for decision-making





Firmware Updates Over The Air (FOTA)

- Enabling remote device updates
- Ensuring security and reliability
- Seamless software delivery